Yiyue Qian

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Summary

Applied Scientist at AWS and ML Tech Lead for GenAI post-training, model customization, agentic systems, and LLM evaluation for AWS AI services. Founding member of the Model Customization team at the AWS Generative AI Innovation Center, with experience translating ambiguous enterprise AI problems into production-ready training, evaluation, and deployment recipes across Amazon Bedrock, Amazon Q, Amazon Nova, and customer production workloads.

Technical Skills

- **Post-Training & Customization** SFT, LoRA/QLoRA, DreamBooth LoRA, RL post-training, synthetic data generation, data filtering, prompt optimization, model evaluation
- **Distributed Training & Inference** FSDP, DeepSpeed, multi-GPU training, H200 training, vLLM, distributed inference, mixed precision, learning-rate scheduling
- **AWS AI Services** Amazon Bedrock, Amazon Nova, Amazon Q, Claude on Bedrock, OpenSearch, S3, AWS GenAI service integration
- **Agentic AI** Strands Agents SDK, MCP, tool orchestration, multi-agent systems, workflow state management, short-term/long-term memory, FastAPI
- **ML & Research** PyTorch, contrastive learning, graph neural networks, hypergraph learning, multimodal learning, vision encoders, RAG
- **Programming** Python, SQL, C++, JavaScript, Linux, AWS

Experience

- **Amazon Web Services, Generative AI Innovation Center** Seattle, USA
  Applied Scientist (ML Tech Lead) *Dec. 2023 - present*
  - **Founding Model Customization Leadership:** Founding member of the Model Customization team at AWS GenAIIC, helping establish post-training, evaluation, and deployment methodologies for adapting foundation models across Amazon Bedrock, Amazon Q, Amazon Nova, and enterprise production workloads.
  - **Amazon Q Model Customization:** Collaborated with Anthropic and Amazon Q teams to fine-tune Claude-family models on real Amazon developer Q&A data; built data augmentation, LLM-as-a-Judge evaluation, Elo-rating evaluation, and metric-specific evaluation pipelines that improved win rate from 13.2% to 71.5% and directly supported the first launch version of Amazon Q.
  - **Amazon Nova VLM Service Enablement:** Led validation of Amazon Nova VLM models through customer engagements and led scientist/MLE teams of up to 5 people to train customer-domain vision encoders for medical X-ray understanding, so that the Nova VLM service can support customer-customized domain vision encoders.
  - **Customer Model Customization:** Led model customization projects converting ambiguous enterprise AI needs into production-ready training and evaluation recipes across Q&A, SQL/query generation with Claude Sonnet 3 customization, Stable Diffusion 3.5 image generation, medical image understanding with DINO adapted to Amazon Nova VLM, and automotive diagnostics with Qwen 3 RL post-training.
  - **Production Agentic AI Systems:** Tech-led production agentic AI systems for financial planning and analysis, supply-chain workflow generation, and enterprise luxury-hotel AI assistants; designed tool hierarchies, workflow state management, retrieval, orchestration, MCP integration, short-term and long-term memory management, and human-review loops.
  - **Technical Leadership:** Led scientist/MLE teams of up to 5 people across 10+ customer-facing and service-facing GenAI workstreams, owning technical direction from data curation and distributed post-training to evaluation, deployment handoff, and customer communication.
  - **Research from Production Problems:** Translated production challenges into research publications on agentic data augmentation, multi-agent LLM-as-a-Judge, AutoData, graph learning, and RL post-training for judge models that distill multi-agent reasoning into single-model inference.
- **Amazon** Seattle, USA
  Applied Scientist Intern *May. 2023 - Sept. 2023; May. 2022 - Aug. 2022*
  - **Graph & Multi-Modal Learning for Marketplace Integrity:** Designed CollusionEmbed and 2TowerFusion for self-supervised graph pretraining and multi-modal collusion detection across buyers, sellers, and products, improving downstream abuse detection AUC by 20–40% over production baselines.
  - **LLM Fine-tuning for Fraud Detection:** Applied prompt engineering and LLM fine-tuning to fraudulent datasets for redirecting-traffic website detection and suspicious product identification.
  - **Ring of Abuser Detection:** Developed and **deployed** a universal multi-modal GNN framework called **U-ROAD** to detect the ring of abusers (e.g., seller-buyer collusion) on *Amazon* over heterogeneous graphs over seller, buyer, and product with multi-modal features (i.e., text and image). (The annual bad debt reduction on Amazon is estimated at $600k.)
  - **Publication: Qian, Y.**, Chen, P., Cui, S., Chen, D.. U-ROAD: Universal Ring-of-Abuser Detection via Multi-Modal Heterogeneous Graph Learning, **KDD** RelKD workshop, 2023.
- **Ping An Technology Co., Ltd.** Shenzhen, China
  Algorithm Engineer in NLP (Full-time) *Jan. 2018 - Aug. 2019*
  - **Model Building:** Built NLP-related models to capture main topics, recognize named entities, analyze relationships among entities, and judge sentiment from unstructured text for Ping An FinTech AI analysis platform.
  - **Risk Event Monitoring:** Developed models on Linux to analyze the topic of global daily financial news such as Reuters News and monitor risk events at different levels (bankruptcy, violation of laws, affairs of executives, etc.). This launched function on the AI analysis platform is estimated to save more than $1 million per year for Ping An Group.

Education

- **University of Notre Dame** — Notre Dame, USA
  PhD in Computer Science — *Jan. 2022 - Dec. 2023*
- **Case Western Reserve University** — Cleveland, USA
  PhD in Computer Science — *Sept. 2019 - Dec. 2021*
- **University College London** — London, UK
  Master in Statistics — *Sept. 2016 - Sept. 2017*

Selected Research

- **Research Focus** Published 20+ papers across NeurIPS, WWW, KDD, ICDM, CIKM, ACL Findings, IJCAI, WSDM, and UAI on LLM evaluation, agentic data systems, graph/hypergraph representation learning, multimodal learning, trustworthy AI, and public-safety AI.
- **Production-to-Research Direction** Research agenda bridges production GenAI challenges and publishable methods, including multi-agent LLM-as-a-Judge, AutoData for open-web data collection, evaluation-driven data augmentation, and RL post-training for judge models.

Selected Publications

- **[Agent, RL] Qian, Y.**, Zhang, S., Song, H. JudgePanel: RL-based Post-Training for Judge Models with Multi-Agent Reasoning, **EMNLP**, 2026. *(Under review)*
- **[RL]** Ma, T., **Qian, Y.**, Li, Y., et al. Non-Monotonic Autoregressive Sequence Model, **ICML**, 2026.
- **[Agent] Qian, Y.**, Ma, T., Zhang, Z., et al. AutoData: A Multi-Agent System for Open Web Data Collection, **NeurIPS**, 2025.
- **[Agent, SFT]** Song, H., Razdan, D., **Qian, Y.**, et al. Learning from Generalization Patterns: An Evaluation-Driven Approach to Enhanced Data Augmentation, **NeurIPS (Workshop)**, 2025.
- **[GNN] Qian, Y.**, Zhang, S., Chen, P., et al. Enhancing E-commerce Representation Learning via Hypergraph Contrastive Learning (RelationEmbed), **WWW**, 2025.
- **[Agent, Judge] Qian, Y.**, Zhang, S.*, Zhou, Y., et al. Enhancing LLM-as-a-Judge via Multi-Agent Collaboration, **AAAI (Workshop)**, 2025.
- **[LLM, GNN]** Ma, T., **Qian, Y.**, Wang, Z., et al. LLM-Empowered Class Imbalanced Graph Prompt Learning for Online Drug Trafficking Detection, **ACL Findings**, 2025.
- **[GNN] Qian, Y.**, Ma, T., Zhang, C., et al. Dual-level Hypergraph Contrastive Learning with Adaptive Temperature Enhancement, **WWW**, 2024.
- **[Multi-Modal Pre-train] Qian, Y.**, Zhang, C., Wen, Q., et al. Co-Modality Graph Contrastive Learning for Imbalanced Node Classification, **NeurIPS**, 2022.

Awards

- Best Paper Award, The Web Conference (WWW) Workshop, 2025
- Best Paper Candidates List, ICDM, 2025
- NeurIPS Scholar Award, 2022
- CIKM Student Travel Award, 2022

Services

- **Conference Senior Program Committee** CIKM (2024–2025).
- **Conference Program Committee** AAAI (2023–2024), CIKM (2023–2024), ICDM (2022–2023), ICLR (2024), ICML (2024), IJCAI (2024), KDD (2022–2025), NeurIPS (2023, 2025), SDM (2024), WSDM (2023–2025), WWW (2023–2025), CIKM (2024–2025)
- **Journal Reviewer** IEEE Transactions on Big Data (2023–2025),
  IEEE Transactions on Dependable and Secure Computing (2023–2025),
  ACM Transactions on Privacy and Security (2023),
  Neurocomputing (2023–2025),
  IEEE Transactions on Neural Networks and Learning Systems (2024–2025),
  Other Journals (2023–2025).
- **Outreach Program Judge** Learning Science Sphere Event at Northpoint Elementary School (2023).
- **Teaching Assistant** Computer Security (CSE-40567, CSDS-344) (2020–2021).